

Wide application of a novel topoisomerase I inhibitor-based drug conjugation technology.

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To establish a novel and widely applicable payload-linker technology for antibody-drug conjugates (ADCs), we have focused our research on applying exatecan mesylate (DX-8951f), a potent topoisomerase I inhibitor, which exhibits extensive antitumor activity as well as significant myelotoxicity, as the payload part. Through this study, we discovered a promising exatecan derivative (DX-8951 derivative, DXd), that has the characteristics of low membrane permeability and shows considerably less myelotoxicity than that shown by exatecan mesylate in an in vitro human colony forming unit-granulocyte macrophage assay. DXd was further used for drug conjugation by using commercially or clinically useful monoclonal antibodies to evaluate the potency of the ADC. The result revealed that the DXd-ADCs targeting CD30, CD33, and CD70 were effective against each of their respective target-expressing tumor cell lines. Moreover, a novel DXd-ADC targeting B7-H3, which is a new target for ADCs, also showed potent antitumor efficacy both in vitro and in vivo. In conclusion, this study showed that this novel topoisomerase I inhibitor-based ADC technology is widely applicable to a diverse number of antibodies and is expected to mitigate myelotoxicity, thereby possibly resulting in better safety profiles than that of existing ADC technologies.

Total Synthesis in Search of Potent Antibody-Drug Conjugate Payloads. From the Fundamentals to the Translational.

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The emergence and evolution of antibody-drug conjugates (ADCs) as targeted cancer therapies in recent years is a living example of the "magic bullet" concept of Paul Ehrlich, introduced by him more than a century ago. Consisting of three components, the antibody serving as the delivery system, the payload drug that kills the cancer cell, and the chemical linker through which the payload is attached to the antibody, ADCs represent a currently hotly pursued paradigm of targeted cancer therapies. While the needed monoclonal antibody falls in the domains of biology and biochemistry, the potent payload and the linker belong to the realm of chemistry. Naturally occurring molecules and their derivatives endowed with high cytotoxic properties have proven to be useful payloads for the first approved ADCs (i.e., Mylotarg, Adcetris, Kadcyca, and Besponsa). The latest approaches and intensifying activities in this new paradigm of cancer therapy demands a variety of payloads with different mechanisms of action in order to address the medical needs for the various types of cancers, challenging synthetic organic chemists to enrich the library of potential payloads. Total synthesis of natural and designed molecules not only provides a powerful avenue to replicate rare naturally occurring compounds in the laboratory but also offers a unique opportunity to rationally design and synthesize analogues thereof for biological evaluation and optimization of ADC payloads. In this Account, we describe our efforts in this area highlighting a number of total synthesis endeavors through which we rendered scarce naturally occurring molecules readily available for biological evaluations and, most importantly, employed the developed synthetic strategies and methods to construct, otherwise unavailable or difficult to reach, designed analogues of these molecules. Specifically, we summarize the total syntheses of natural and designed molecules of the calicheamicin, uncialamycin, tubulysin, trioxacarcin, epothilone, shishijimicin, namenamicin, thailanstatin, and disorazole families of compounds and demonstrate how these studies led to the discovery of analogues of higher potencies, yet some of them possessing lower complexities than their parent

compounds as potential ADC payloads. The highlighted examples showcase the continuing impact of total synthesis of natural products and their analogues on modern medicine, including the so-called biologics and should prove useful and inspirational in advancing both the fields of total synthesis and biomedical research and the drug discovery and development process.

UPLC-based assay to assess the hydrophobicity of Antibody-Drug Conjugate (ADC) payloads.

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Antibody-Drug Conjugates (ADCs) consist of antibodies attached to cytotoxic small molecules or biological agents (i.e., payloads) through chemical linkers which may be cleavable or non-cleavable. The development of new ADCs is challenging, particularly the process of attaching the linker-payload construct to the antibody (i.e., the conjugation process). One of the major problems associated with conjugation is high hydrophobicity of the payload which can lead to low yields of the ADC through aggregation and/or lower than desired Drug-Antibody Ratios (DARs). We report here a UPLC-based assay that can be used to study the physicochemical properties of ADC payloads at an early stage of development, and to provide information on whether the hydrophilic-hydrophobic balance is suitable for conjugation or further physicochemical optimization is required. The assay is relatively simple to establish and should be of use to those working in the ADC area.

Antibody-drug conjugates as targeted therapies: Are we there yet? A critical review of the current clinical landscape.

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Antibody-drug conjugates (ADCs) are targeted therapies with the expectation of broadened therapeutic window due to tumor-specific drug delivery. Recent approvals, including ADCs with a novel payload class, topoisomerase-1 inhibitors, generated renewed excitement in the field. We provide a critical review of approved and late-stage molecules, discuss strategies in solid tumors and ADCs outside oncology. Our pharmacokinetics-based assessment of targeting suggests that ADCs, especially in solid tumors, rely on additional mechanisms for efficacy including slow-release of the payload to the circulation at potentially efficacious levels. Further adjustments in the technology are needed to fulfill the promise of true targeted drug delivery.

Antibody-Drug Conjugates: A Review of Approved Drugs and Their Clinical Level of Evidence.

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Antibody-drug conjugates (ADCs) are an innovative family of agents assembled through linking cytotoxic drugs (payloads) covalently to monoclonal antibodies (mAbs) to be delivered to tumor tissue that express their particular antigen, with the theoretical advantage of an augmented therapeutic ratio. As of June

2023, eleven ADCs have been approved by the Food and Drug Administration (FDA) and are on the market. These drugs have been added to the therapeutic armamentarium of acute myeloblastic and lymphoblastic leukemias, various types of lymphoma, breast, gastric or gastroesophageal junction, lung, urothelial, cervical, and ovarian cancers. They have proven to deliver more potent and effective anti-tumor activities than standard practice in a wide variety of indications. In addition to targeting antigen-expressing tumor cells, bystander effects have been engineered to extend cytotoxic killing to low-antigen-expressing or negative tumor cells in the heterogenous tumor milieu. Inevitably, myelosuppression is a common side effect with most of the ADCs due to the effects of the cytotoxic payload. Also, other unique side effects are specific to the tissue antigen that is targeted for, such as the cardiac toxicity with Her-2 targeting ADCs, and the hemorrhagic side effects with the tissue factor (TF) targeting Tisotumab vedotin. Further exciting developments are centered in the strategies to improve the tolerability and efficacy of the ADCs to improve the therapeutic window; as well as the development of novel payloads including (1) peptide-drug conjugates (PDCs), with the peptide replacing the monoclonal antibody, rendering greater tumor penetration; (2) immune-stimulating antibody conjugates (ISACs), which upon conjugation of the antigen, cause an influx of pro-inflammatory cytokines to activate dendritic cells and harness an anti-tumor T-cell response; and (3) the use of radioactive isotopes as a payload to enhance cytotoxic activity.

Antibody Drug Conjugate bioinformatics: drug delivery through the letterbox.

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Antibodies appear to be the first line of defence in the adaptive immune response of vertebrates and thereby are involved in a multitude of biochemical mechanisms, such as regulation of infection, autoimmunity, and cancer. It goes without saying that a full understanding of antibody function is required for the development of novel antibody-interacting drugs. These drugs are the Antibody Drug Conjugates (ADCs), which are a new type of targeted therapy, used for example for cancer. They consist of an antibody (or antibody fragment such as a single-chain variable fragment [scFv]) linked to a payload drug (often cytotoxic). Because of the targeting, the side effects should be lower and give a wider therapeutic window. Overall, the underlying principle of ADCs is to discern the delivery of a drug that is cytotoxic to a target that is cancerous, hoping to increase the antitumoural potency of the original drug by reducing adverse effects and side effects, such as toxicity of the cancer target. This is a pioneering field that employs state-of-the-art computational and molecular biology methods in the fight against cancer using ADCs.

In vivo safety testing of Antibody Drug Conjugates.

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Antibody Drug Conjugates (ADCs) are complex multi-domain biotherapeutics which combine, with the aid of a chemical linker, tumor-targeting antibodies with potent small molecule cytotoxicants (also called warhead or payload) for the treatment of cancer. ADCs are a rapidly growing class of pharmaceuticals with nine FDA-approved drugs already on the market and over eighty at different stages of clinical development, and also an increasing number under evaluation for non-oncological indications. Off-target toxicity and a narrow therapeutic index has been a problem with ADCs. This has driven the search for better targeting (disease models, cell surface antigens), linker stability, and payload specificity. Analysis

of regulatory approval documents, scientific publications and ICH guidance shows that safety evaluation of ADCs requires novel integrated strategies different from both standard chemotherapy and antibody-based products, e.g. development and validation of ADC analytical assays. There is no ADC-specific guidance on safety evaluation; current guidance emphasises the need for an adaptive approach but more ADC-specific guidance is now arguably possible. The data now available will help to optimize primary target specificity, select appropriate combination partners, develop in silico models, and provide guidance for preclinical and clinical safety evaluation for the next generation of this class of multi-domain therapeutics.

Boronic acids as building blocks for the construction of therapeutically useful bioconjugates.

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Bioconjugates are multifunctional constructs in which biomolecules like peptides, proteins, vitamins and nucleic acids are endowed with the properties of specific payloads. These constructs recently emerged as a new generation of high-precision therapeutics, with several representatives reaching the market. This success stimulated an intense search for new biocompatible synthetic methodologies to connect both components and to control the bioconjugate's function. Despite the remarkable advances made in this field, most of the technologies developed for the construction of bioconjugates were engineered to yield stable constructs that can endure complex physiological conditions. Because of this, the use of reversible covalent bonds in the synthesis of bioconjugates has been rather overlooked, notwithstanding the potential of this strategy to generate stimuli responsive constructs that may operate in areas like the selective delivery of drugs, live-cell imaging and new theranostic approaches. Boronic acids are a well-known class of reagents that have been widely used in modern synthesis for the formation of C-C and C-heteroatom bonds. Apart from this, boronic acids exhibit an exquisite reversible coordination profile that can be explored as a molecular construction tool featuring specific mechanisms to control the structure and biological properties of bioconjugates. In this review, the use of boronic acids in the construction of therapeutically useful bioconjugates will be discussed, focusing on the molecular mechanisms that allow the use of these reagents as bioconjugation warheads, as central pieces of linker structures and as functional payloads.

Immune biology of glioma-associated macrophages and microglia: functional and therapeutic implications.

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CNS immune defenses are marshaled and dominated by brain resident macrophages and microglia, which are the innate immune sentinels and frontline host immune barriers against various pathogenic insults. These myeloid lineage cells are the predominant immune population in gliomas and can constitute up to 30-50% of the total cellular composition. Parenchymal microglial cells and recruited monocyte-derived macrophages from the periphery exhibit disease-specific phenotypic characteristics with spatial and temporal distinctions and are heterogeneous subpopulations based on their molecular signatures. A preponderance of myeloid over lymphoid lineage cells during CNS inflammation, including gliomas, is a contrasting feature of brain immunity relative to peripheral immunity. Herein we discuss

glioma-associated macrophage and microglia immune biology in the context of their identity, molecular drivers of recruitment, nomenclature and functional paradoxes, therapeutic reprogramming and polarization strategies, relevant challenges, and our perspectives on therapeutic modulation.

Nanoparticles Hitchhike on Monocytes for Glioblastoma Treatment after Low-Dose Radiotherapy.

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Glioblastomas (GBMs) are aggressive primary brain tumors with fatal outcome. Traditional chemo-radiotherapy has poor therapeutic effect and significant side effects, due to the drug and radiotherapy (RT) resistance, natural blood-brain barrier, and high-dose RT damage. Even more, tumor-associated monocytes (macrophages and microglia, TAMs) constitute up to 30%-50% of the GBM cellular content, and the tumor microenvironment (TME) in GBM is extremely immunosuppressive. Here, we synthesized nanoparticles (D@MLL) that hitchhike on circulating monocytes to target intracranial GBMs with the assistance of low-dose RT. The chemical construction of D@MLL was DOX·HCl loaded MMP-2 peptide-liposome, which could target monocytes by the surface modified lipoteichoic acid. First, low-dose RT at the tumor site increases monocyte chemotaxis and induces M1 type polarization of TAMs. Subsequently, the intravenous injected D@MLL targets circulating monocytes and hitchhikes with them to the central site of the GBM area. DOX·HCl was then released by the MMP-2 response, inducing immunogenic cell death, releasing calreticulin and high-mobility group box 1. This further contributed to TAMs M1-type polarization, dendritic cell maturation, and T cell activation. This study demonstrates the therapeutic advantages of D@MLL delivered by endogenous monocytes to GBM sites after low-dose RT, and it provides a high-precision treatment for GBMs.
